DAY 21 – Final Integration and Testing

Day 21 covered the final and most comprehensive stage: **integrating all steps into a complete end-to-end forecasting workflow**. This included data preprocessing, modeling, visualization, evaluation, and exporting — creating a complete, ready-to-present project.

🔹 1. Combining All Components

The final workflow included:

• Data loading and cleaning
• EDA and visualization
• Model training using Prophet
• Prediction generation
• Error evaluation
• Result exporting

Integration ensures smooth execution from start to finish.

🔸 2. Code Structuring and Organization

We organized code into:

✓ Data preprocessing section
✓ Visualization block
✓ Model training setup
✓ Forecast generation cell
✓ Evaluation metrics cell
✓ Exporting and saving section

This makes the notebook professional and easy to maintain.

🔸 3. Final Testing of Workflow

The workflow was tested for:

• Error-free execution
• Correct output generation
• Proper visuals and predictions
• Accuracy of results
• Reproducibility

Testing ensures the project works as expected every time.

🔸 4. Project Review and Validation

We reviewed:

• Cleaned dataset
• Prophet model predictions
• Error and residual plots
• Forecast accuracy
• Final exported files

Everything was checked to ensure completeness and correctness.

🔸 5. Completion of the Training Project

This final step marked:

• The successful implementation of a real forecasting model
• A full understanding of ML workflow
• Practical experience with Python, Pandas, NumPy, Plotly, and Prophet
• A strong foundation in data science fundamentals

It completed the training journey with a fully functional project.

Key Takeaways from Day 21

• Integration unifies the entire project
• Testing ensures smooth execution
• Proper structuring improves clarity
• Final review completes the workflow
• This marks the completion of a full data science mini-project